Chapter 1-6

QBIO 305 FA26

Summary of lecture slides

Topics

- Sampling bias (ascertainment bias)
  - Some members of the population are less likely to be included
- Simple random sample
  - Every member of the population has same chance of being included in the sample (representative)
  - Members are chosen independently of each other
  - Representative sample
- Non-simple random samples can still be useful

Case control studies

- Take a sample with the trait you’re trying to study (cases)
- Take another sample without the trait
  - Controls are selected to match cases for other characteristics such as age, income, etc.

Randomized control experiment

- Randomly assign patients to each group
  - Confounding variables should approximately balance between groups
- Double-blind
  - Patients shouldn’t know which group they’re in
  - Researchers shouldn’t know which patients are getting a drug or placebo
- Researchers are doing an experiment

Chapter 2

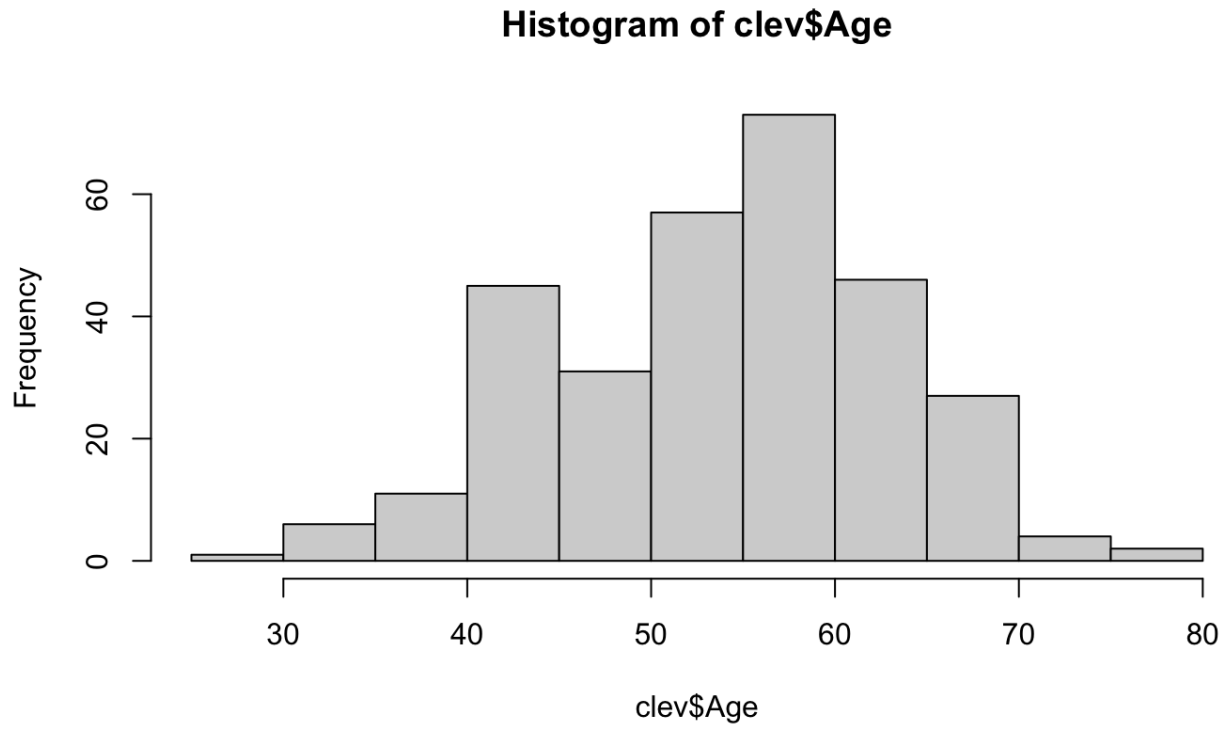
- Variables
  - Quantitative: discrete, continuous
  - Qualitative: categorical, ordinal
- Median
- Mean
- IQR
- 5 number summary
  - Minimum, Q1, median, Q3, maximum

Observational unit

- Sometimes n persons are sampled and multiple variables are collected for each
  - E.g. blood type, gender, and weight of 300 babies are collected
  - The baby is called the “observational unit”

Bar charts/histograms

- Area of one or several bars is proportional to the corresponding frequency



Grouped frequency distribution

- Continuous variables require a decision of how to group the data
- E.g. grouping weight for children:
  - The distribution for 5 lb bins may look different than 10 lb bins

Distribution shapes

- Bell-shaped (or Gaussian or normal)
- Bimodal (2 peaks)
- Asymmetric
  - Skewed to the right (outlier is on right side)
  - Skewed to the left
- Exponential

Descriptive statistics

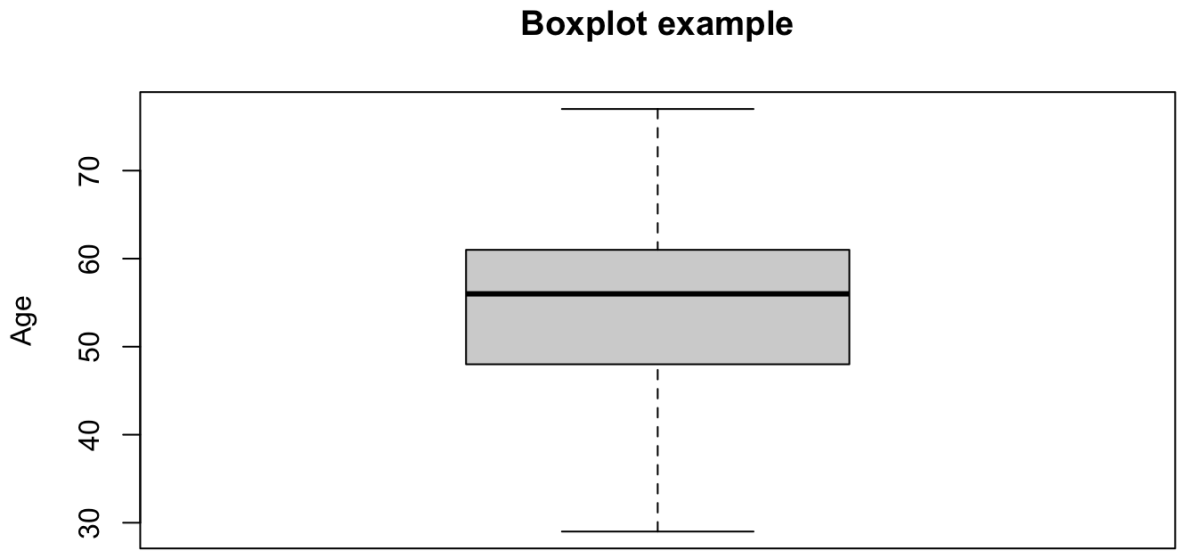
- Median
  - The value that most nearly lies in the middle of the sample
    - E.g. 5, 10, 15 → median is 10
  - If the sample size is even, we average the 2 middle numbers
    - E.g. 5, 10, 15, 15 → median is 12.5
- Mean
  - The sum of the samples divided by the sample size
    - E.g. 5, 10, 15, 15 → mean is 11.25

Robustness

- A statistic is robust if it is relatively unaffected by changes in a small portion of the data
- The **median** is **robust** and the **mean** is **not**
- Trimmed mean:
  - Removing a percentage of the highest and lowest data points before calculating the mean

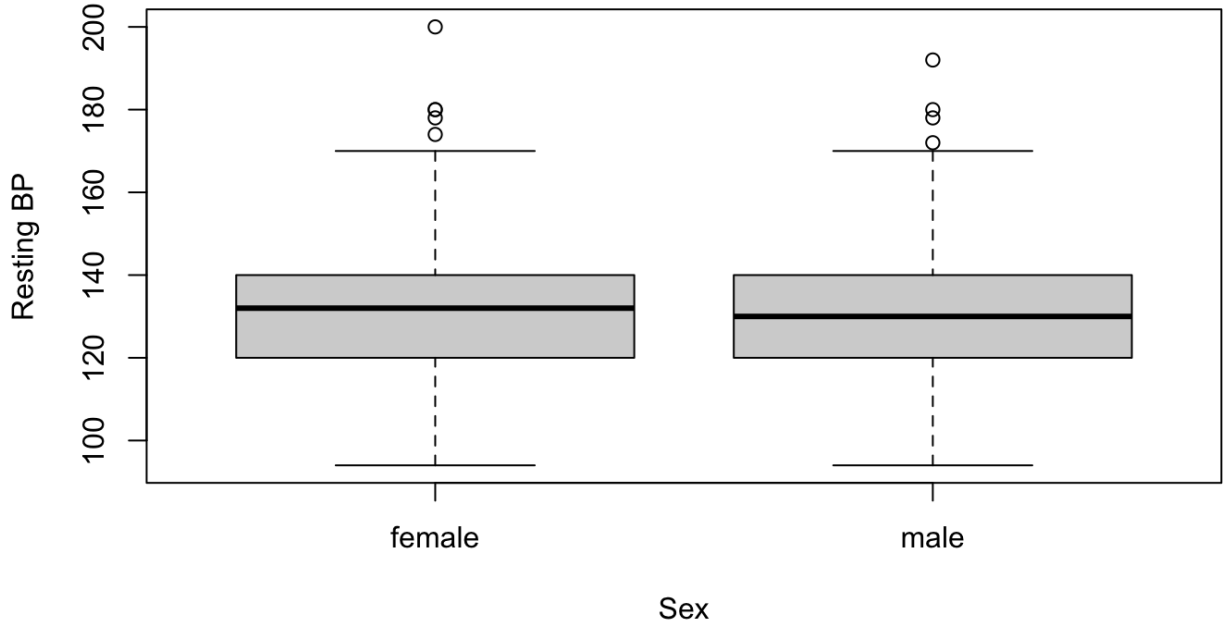
Boxplots

- Median splits distribution into two parts (thick line, lower and upper)
- The quartiles split each of the parts in half
  - The first quartile (Q1, bottom of box) splits the lower half
  - The third quartile (Q3, top of box) splits the upper half
- Interquartile range (IQR)
  - The difference between the third and first quartiles
  - This measures the spread of the middle 50% of the distribution
- Boxplot with no outliers:
  - 5 number summary: minimum, Q1, median, Q3, maximum



Boxplots - outliers

- Box plot with outliers:
  - Lower fence = Q1 – (1.5 x IQR)
  - Upper fence = Q3 + (1.5 x IQR)
  - Any data point lower than the lower fence or higher than the upper fence is an outlier
  - Whiskers extend to smallest and largest data points that are not outliers
  - Outliers are identified by dots



Standard deviation

$$s = \sqrt{\frac{\sum_i (y_i - \bar{y})^2}{n - 1}}$$

Where $\bar{y}$ is the mean

Linear transformations a+bX

- Median/Mean
  - If we add a number, add it to the median
  - If we multiply by a number, multiply the median by that number
  - Also applies to Q1 and Q3
  - IQR/sd: only the multiplication affects this statistic, the addition cancels
- For a sample, if the mean is 25 and standard deviation is 5, what would the mean and standard deviation be if we multiply each sample by 3 and then add 5
  - Y = 3x+5
    - Mean will be 3*25+5 = 80
    - Standard deviation is 15

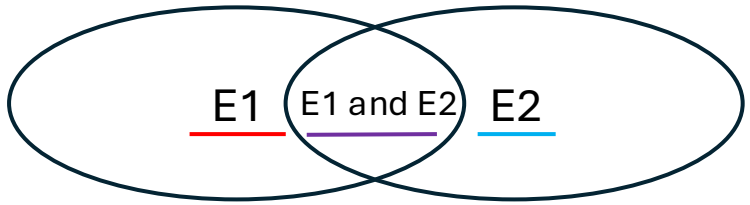
Probability

- A probability is a numerical quantity that expresses the likelihood of an event
- The probability of an event E is written Pr{E} or P(E)
- Pr{E} is always between 0 and 1 (inclusive)

Probability rules

- Probability of an event is always between 0 and 1 (inclusive)
- The probability of all possible events is 1
- The probability an event does not happen is one minus the probability that it does happen
  - The complement of E (denoted E^c) is the even that E does not happen, $Pr\{E^c\} = 1 - Pr\{E\}$
- For any two events E1 and E2,

  $Pr\{E1 \textbf{ or } E2\} = Pr\{E1\} + Pr\{E2\} - Pr\{E1 \text{ and } E2\}$
  - True even if $Pr\{E1 \text{ and } E2\} = 0$



Disjoint

- Two events E1 and E2 are said to be disjoint (or mutually exclusive) if Pr{E1 and E2} = 0
- Disjoint means ‘no overlap’
  - E.g. low income, medium income, and high income would be disjoint stressed and not stressed are disjoint

	Income			
	Low	Medium	High	Total
Stressed	526	274	216	1,016
Not stressed	1,954	1,680	1,899	5,533
Total	2,480	1,954	2,115	6,549

Conditional probability

- The conditional probability of E2 given E1 is:

$$\Pr(E_2|E_1) = \frac{\Pr(E_2 \; and \; E_1)}{\Pr(E_1)}$$

Conditional probability

$$\Pr(E_2|E_1) = \frac{\Pr(E_2 \; and \; E_1)}{\Pr(E_1)}$$

- Pr(med income|stressed) = Pr(med income and stressed)/Pr(stressed)
  = (274/6549)/(1016/6549)
  = 274/1016 = ~.27

	Income			
	Low	Medium	High	Total
Stressed	526	274	216	1,016
Not stressed	1,954	1,680	1,899	5,533
Total	2,480	1,954	2,115	6,549

Conditional probability

$$\Pr(E_2|E_1) = \frac{\Pr(E_2 \; and \; E_1)}{\Pr(E_1)}$$

- Pr(stressed) = 0.08+0.04+0.03=0.15
- Pr(med income|stressed) = Pr(med income and stressed)/Pr(stressed)
  = 0.04/0.15 = ~.27

	low	**med**	**high**
stressed	0.08	0.04	0.03
not stressed	0.30	0.26	0.29

	Income			
	Low	Medium	High	Total
Stressed	526	274	216	1,016
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Total	2,480	1,954	2,115	6,549

Multiplication rule

- For any two events E1 and E2,

$$\Pr\{E1 \text{ and } E2\} = \Pr\{E1\} \times \Pr\{E2|E1\}$$

Multiplication rule

- For any two events E1 and E2,

$$\Pr\{E1 \text{ and } E2\} = \Pr\{E1\} \times \Pr\{E2|E1\}$$

$$\Pr(\text{low inc. and not stressed}) = \Pr(\text{low inc.}) \times \Pr(\text{not stressed} \mid \text{low inc.})$$

$$= (2480/6549) \times (1954/2480)$$

$$= 1954/6549$$

	Income			
	Low	Medium	High	Total
Stressed		274	216	1,016
Not stressed	1,954	1,680	1,899	5,533
Total	2,480	1,954	2,115	6,549

Note

- Pr{E1|E2} =/= Pr{E2|E1}
- E.g.
  - Pr(high inc | stressed) = 216/1016
  - Pr(stressed | high inc) = 216/2115

	Income			
	Low	Medium	High	Total
Stressed	526	274	216	1,016
Not stressed	1,954	1,680	1,899	5,533
Total	2,480	1,954	2,115	6,549

Independence

- Events E1 and E2 are independent if Pr{E1 and E2} = Pr{E1} x Pr{E2}
- This is equivalent to Pr{E2|E1} = Pr{E2}
- Also equivalent to Pr{E1|E2} = Pr{E1}
- Why? two events are independent if knowing that one event occurred does not change the probability of the other event occurring
- Showing any of the 3 is not true is sufficient!

Law of total probability

- For any two events E and F,

$$P(F) = P(E \text{ and } F) + P(E^c \text{ and } F)$$

- Using the multiplication rule,

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c)$$

$$P(\text{low income}) = P(\text{low}|\text{stress})P(\text{stress}) + P(\text{low}|\text{not stress})P(\text{not stress})$$

Law of total probability

- For any two events E and F,

$$P(F) = P(E \text{ and } F) + P(E^c \text{ and } F)$$

- Say event F = not stressed, E = medium income

P(not stressed) = P(med inc and not stressed) + P(not medium inc. + not stressed)

= P(med inc and not stressed) + P(low inc + not stressed) + P(high inc + not stressed)

= (1680/6549) + (1954/6549) + (1899/6549)

= 5533/6549

	low	med	high
stressed	0.08	0.04	0.03
not stressed	0.30	0.26	0.29

	Income			
	Low	Medium	High	Total
Stressed	526	274	216	1,016
Not stressed	1,954	1,680	1,899	5,533
Total	2,480	1,954	2,115	6,549

Bayes' Theorem

- For any two events E and F,

$$P(E|F) = P(F|E)P(E)/P(F)$$

- With the law of total probability,

$$P(E|F) = P(F|E)P(E)/(P(F|E)P(E) + P(F|E^c)P(E^c))$$

Bayes' Theorem

- For any two events E and F,

$$P(E|F) = P(F|E)P(E)/P(F)$$

P(stressed | med income) = P(med income|stressed)P(stressed)/P(med income)

= (.2667)(.15)/.3 = .13333

We can check using the conditional probability from before:

P(stressed | med income) = P(stressed and med)/P(med) = .04/.3 = .13333

	low	med	high
stressed	0.08	0.04	0.03
not stressed	0.30	0.26	0.29

Random variable

- Variable that takes on numerical values that depend on the outcome of a chance operation
- Discrete random variable
  - The sum of all the histogram heights is 1
- Continuous random variable
  - The integral under the density curve is 1

Mean of a random variable

- Also known as the expected value or expectation,

$$\mu_Y = E(Y) = \sum_i y_i P(Y = y_i)$$

- The mean is about the average of the random variable distribution
- The **mean of a random variable** is different from the **sample mean**
  - Different samples have different sample means but the random variable mean will remain the same (e.g. rolling a die)

Variance and standard deviation of a random variable

$$\sigma_Y^2 = Var(Y) = \sum_i (y_i - \mu_Y)^2 P(Y = y_i)$$

- Standard deviation (σ_Y) is the square root of the variance
- The variance and standard deviation quantify the spread (or dispersion) of the distribution

Mean and Variance of continuous RV

$$\mu = \int x f(x) dx$$

$$\sigma^2 = \int (x - \mu)^2 f(x) dx$$

A few more things…

- Consider two random variables:
  - X has mean μ_X and variance σ_X^2
  - Y has mean μ_Y and variance σ_Y^2
- Mean of a + X: a + μ_X
- Mean of bX: $b\mu_X$
- Mean of X+Y: $\mu_X + \mu_Y$
- Mean of X-Y: $\mu_X - \mu_Y$
- Variance of a+bX: $b^2\sigma_X^2$
  - Standard deviation of a+bX: $b\sigma_X$
- Variance of X+Y (assuming independence): $\sigma_X^2 + \sigma_Y^2$
- Variance of X-Y (assuming independence): $\sigma_X^2 + \sigma_Y^2$

Sample statistics

- For constants a and b:

	Y	a+bY
Mean	$\bar{y}$	$a + b\bar{y}$
Median	m	a+bm
Standard deviation	s	bs
IQR	d	bd

Random variables X and Y

- Where
  - X has mean μ_X and variance σ_X^2
  - Y has mean μ_Y and variance σ_Y^2
  - a and b are constants:

	a+bX	X + Y	X – Y
Mean	$a + b\mu_X$	$\mu_X + \mu_Y$	$\mu_X - \mu_Y$
Variance	$b^2\sigma_X^2$	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.	$\sigma_X^2 + \sigma_Y^2$ if X and Y indep.

Bernoulli RV (a.k.a. biased coin)

- p = probability of success (heads)
  - P(Y=1) = p
- 1-p = probability of failure (tails)
  - P(Y=0) = 1-p
- E(Y) = p
- Var(Y) =p(1-p)
- $\sigma = \sqrt{p(1-p)}$

Bernoulli RV (a.k.a. biased coin)

- If the odds of getting a success is .8, what is the variance?
  - Var(Y) = p(1-p) = .8(.2) = .16
  - What's the standard deviation? .4

Binomial random variable

- If there are n independent trials (with success probability of p), count the number of successes
  - Or how many heads when flipping a weighted coin
  - This is the same as the sum of n Bernoulli random variables

$$Z = \sum_{i=1}^{n} Y_i \, , where \; each \; Y_i \; is \; an \; independent \; Bernoulli \; RV$$

$$E(Z) = np$$

$$Var(Z) = np(1-p)$$

$$\sigma = \sqrt{np(1-p)}$$

Binomial formula

- $P(exactly\ j\ successes) = \binom{n}{j} p^j (1-p)^{n-j}$
- $\binom{n}{j} = \frac{n!}{j!(n-j)!}$

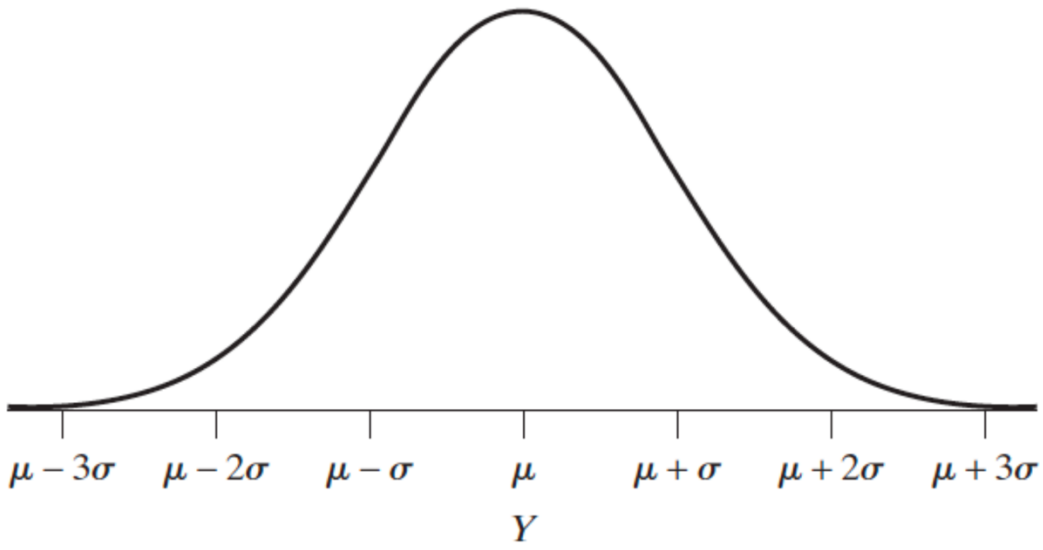
- This makes calculating exact number of successes easy
- “j or more successes” and “j or less successes” is still difficult
- Approximation for this later on

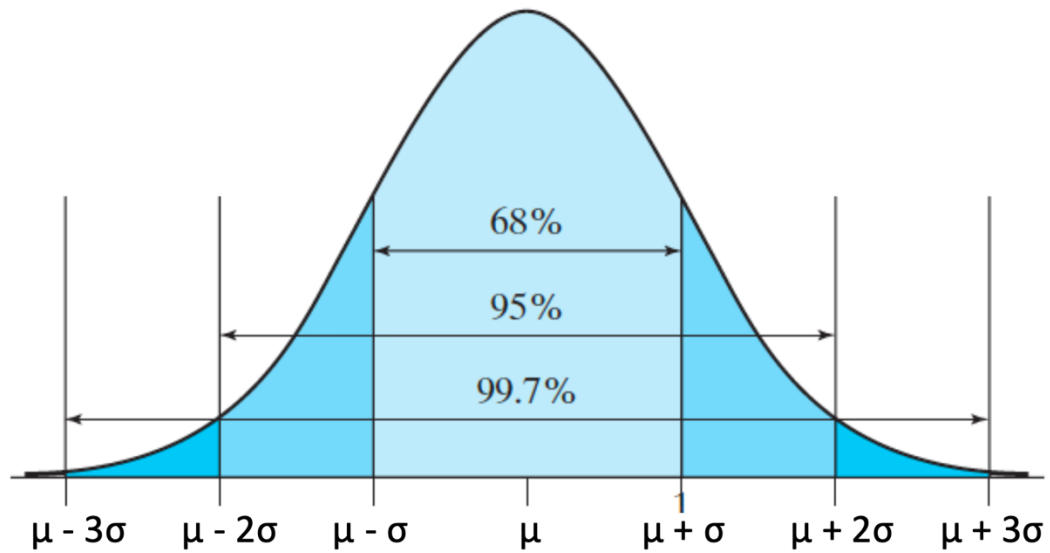
Binomial examples

- Suppose p=.25, n=40
  - What are the odds there is exactly 2 successes?
    - $\binom{n}{j} p^j (1-p)^{n-j} = \binom{40}{2} .25^2 (.75)^{38}$
  - What are the odds there are no successes?
    - $\binom{n}{j} p^j (1-p)^{n-j} = \binom{40}{0} .25^0 (.75)^{40}$
  - What are the odds there is at least one success?
    - P(at least one success) = 1 – P(no successes) = 1 - $\binom{40}{0} .25^0 (.75)^{40}$

Normal distribution

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$$

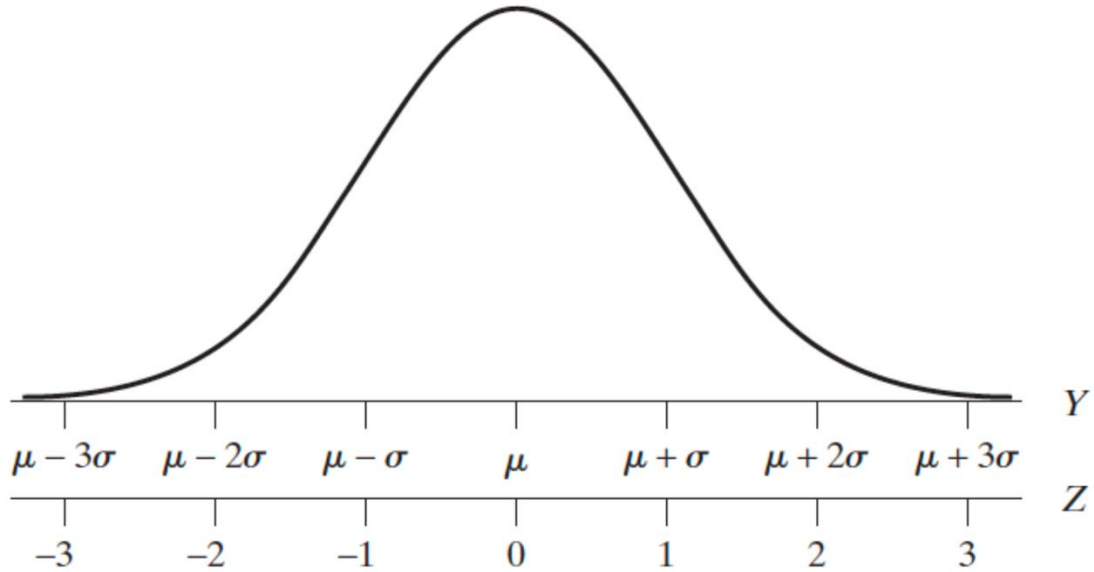



68%
95%
99.7%
μ - 3σ
μ - 2σ
μ - σ
μ
μ + σ
μ + 2σ
μ + 3σ

Normal distribution properties

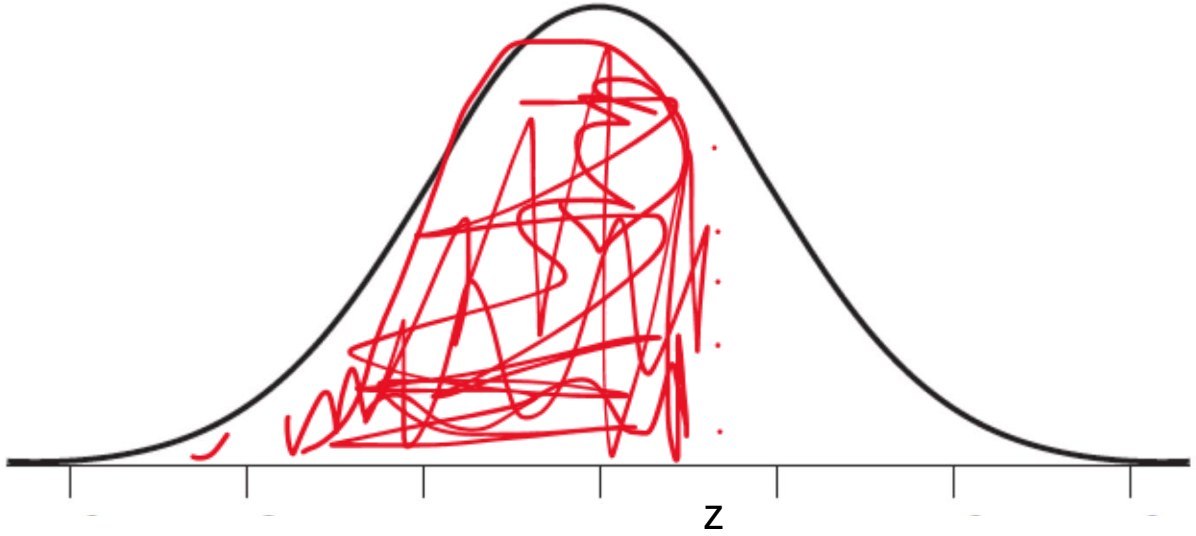
- Assume Y is normal with mean μ and SD σ
  - Y can be written as N(μ, σ)
- Then we can define $\mathrm{Z} = \frac{Y-\mu}{\sigma}$
  - Z is also normal with mean 0 and SD 1
  - In other words, Z is N(0,1)
  - This is called standard normal

Standardized Scale

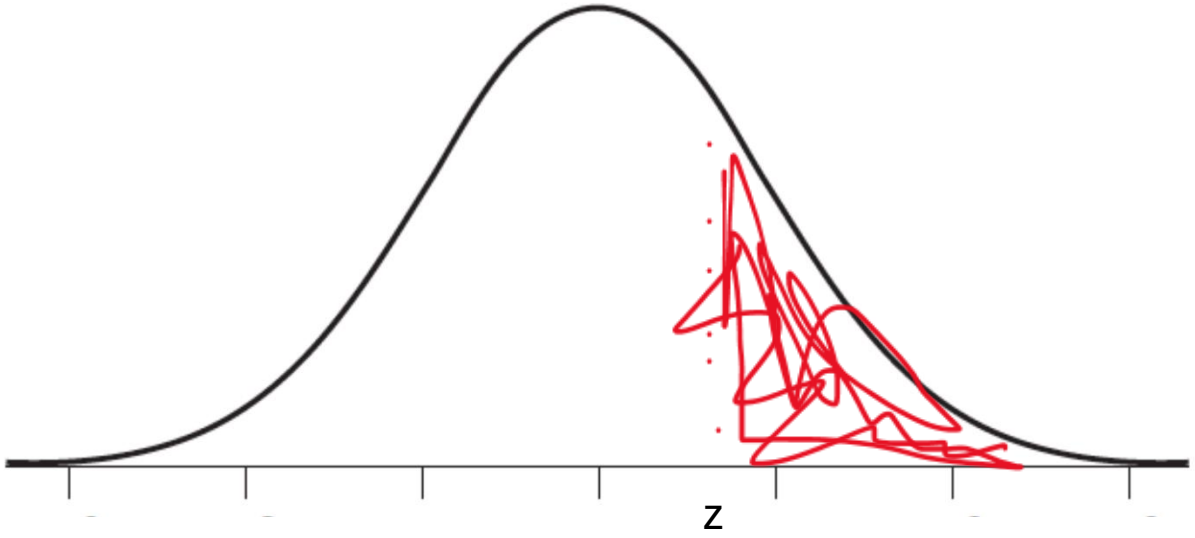


Standardization Formula

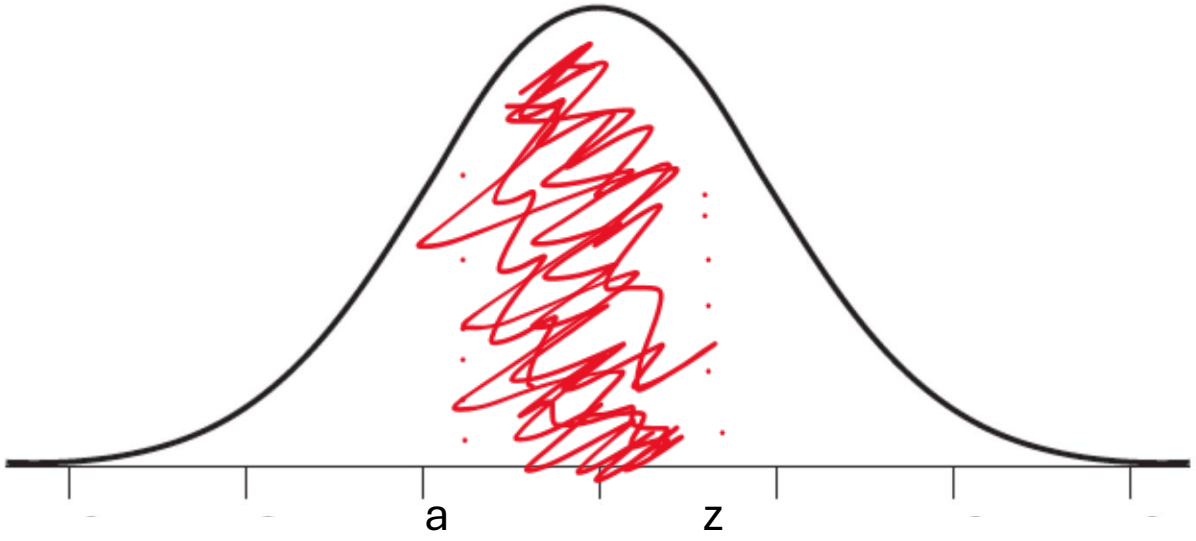
$$Z = \frac{Y - \mu}{\sigma}$$



Probability of z

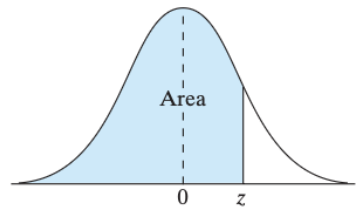


1 – probability of z



Probability of z – probability of a

Table 3 Areas Under the Normal Curve



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table 3 Areas Under the Normal Curve *(continued)*

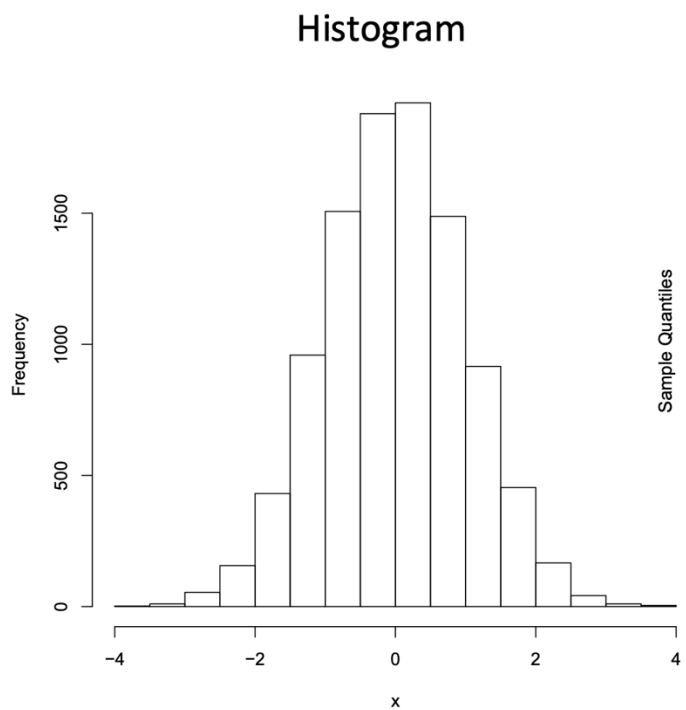
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

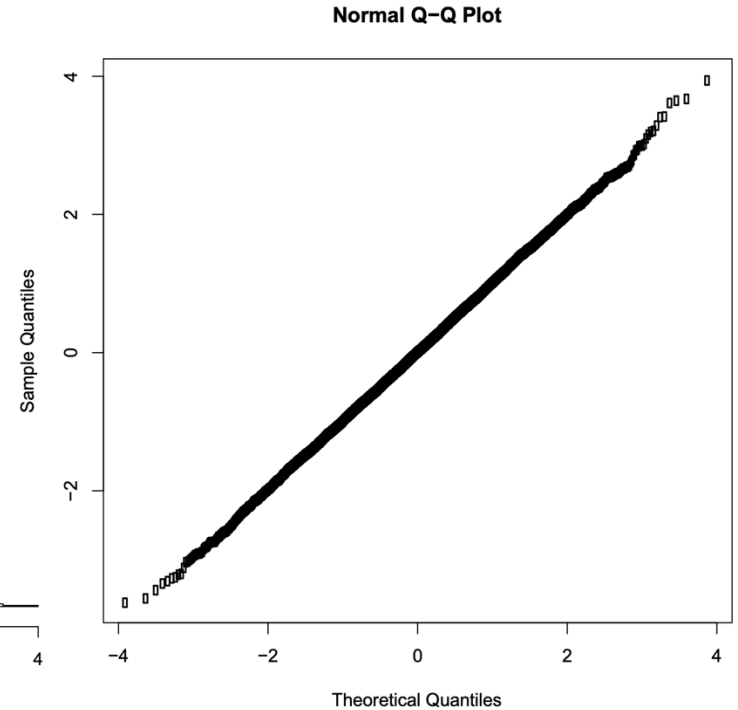
Fish example

- Population of herring is normally distributed
  - Mean 54mm and SD 4.5mm
  - What fraction of fish are less than 60mm long?
    - $\mathrm{Z} = \frac{Y-\mu}{\sigma} = \frac{60-54}{4.5} = 1.33$
    - P(Z<1.33)
  - What fraction of fish are greater than 51mm long?
    - $\mathrm{Z} = \frac{Y-\mu}{\sigma} = \frac{51-54}{4.5} = -0.67$
    - P(Z>-0.67) = 1-P(Z<-0.67) = …

QQ plots

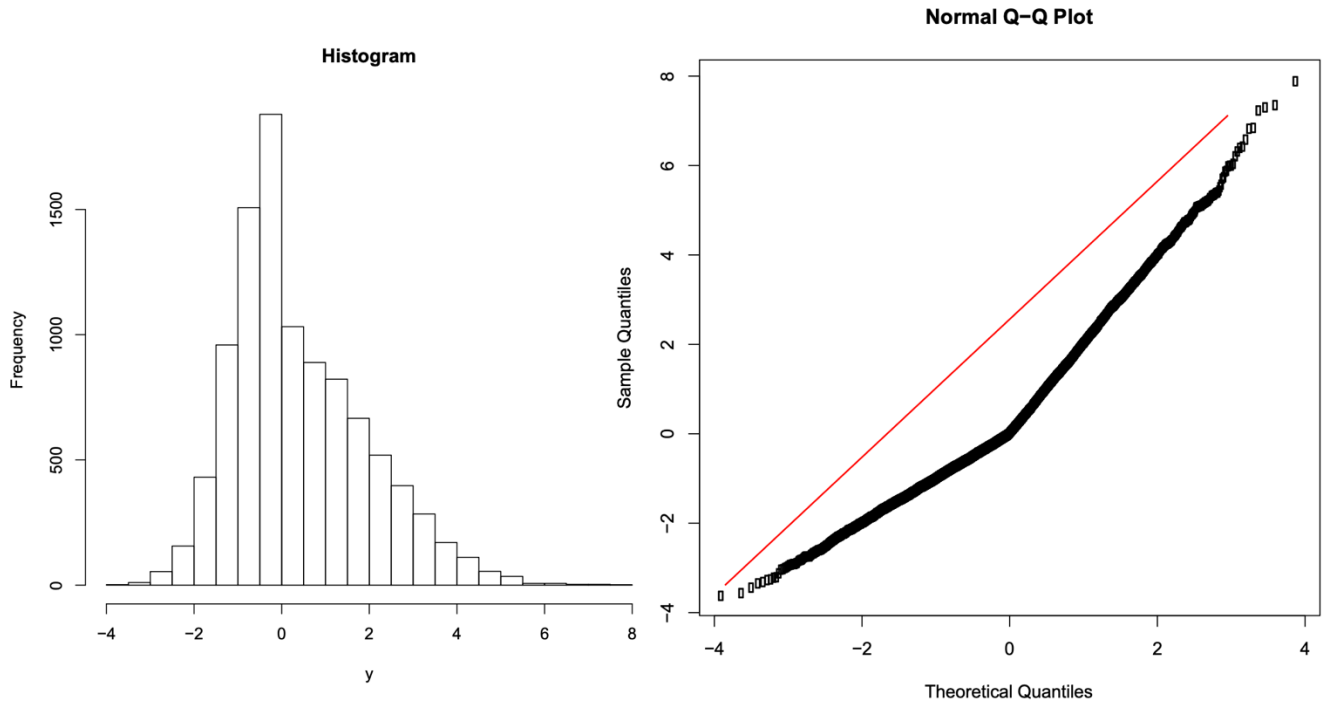
- Test for normality
- Many things are normally distributed
- Order the data from in increasing order
- Assume data is normally distributed and can calculate expected value based on quantile





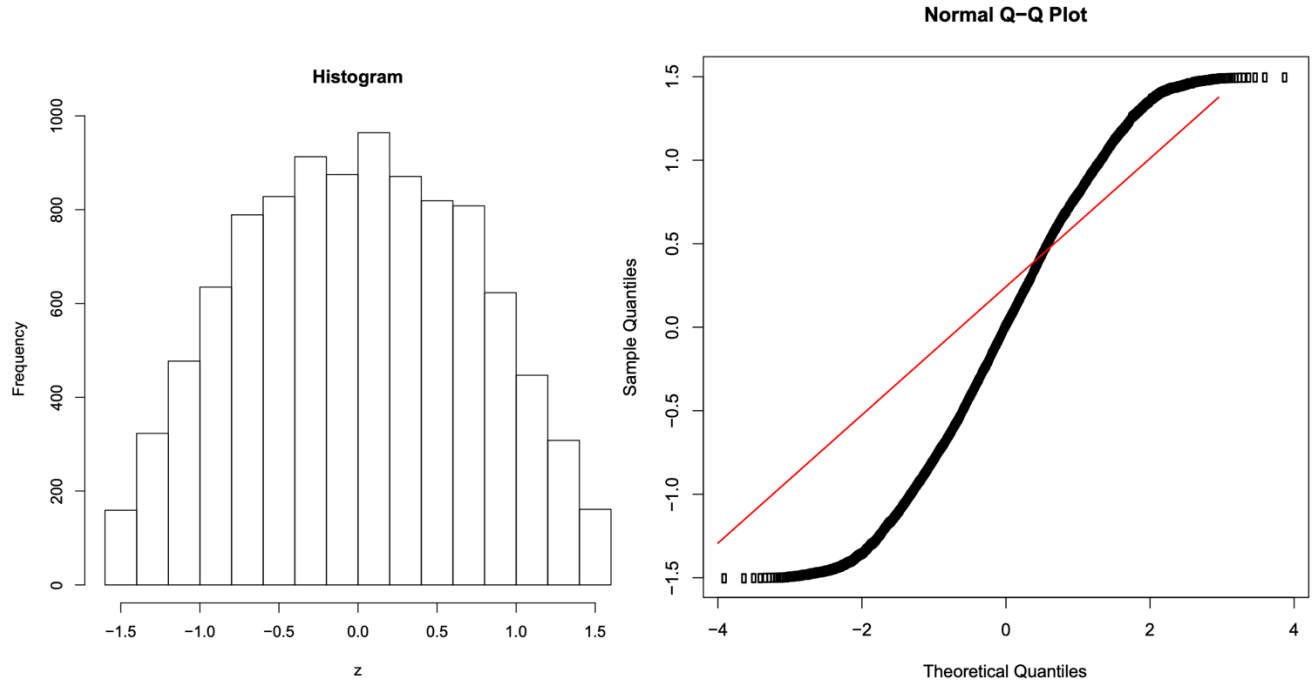
Straight line implies the distribution is normal

Not straight line implies the distribution is not normal

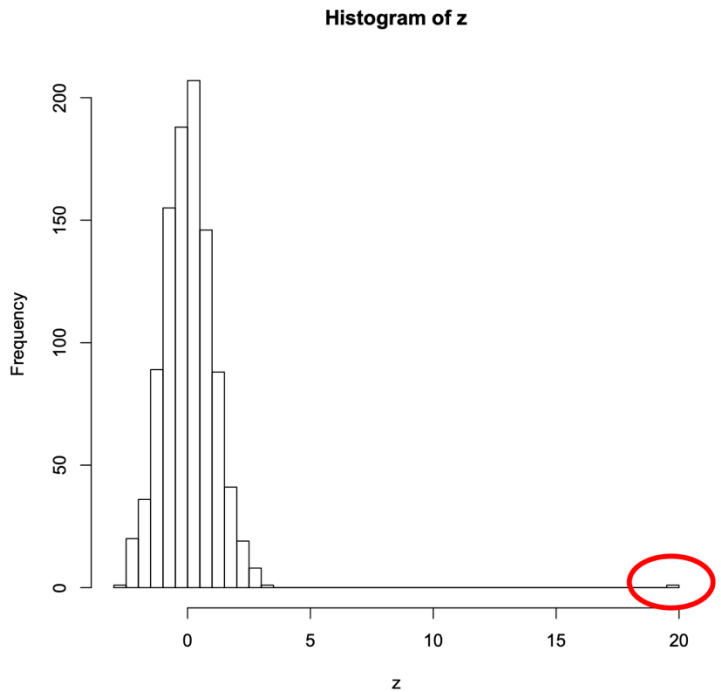


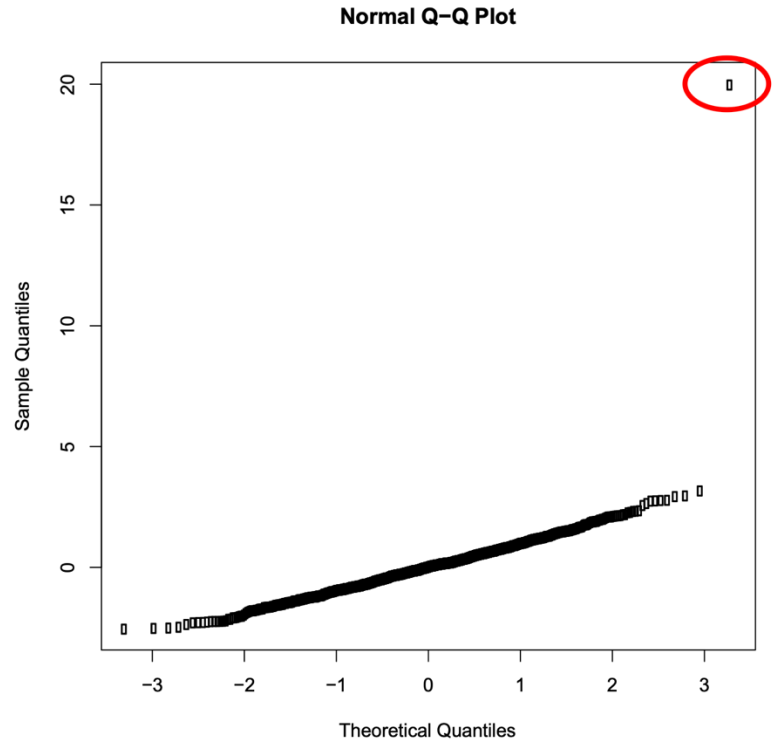
Not straight (I added the red line for comparison)

So not normal



Not straight so not normal





This outlier makes the distribution not normal

Reminder: Simple random sample

- Definition
  - Every member of the population has the same chance of being included (sample is representative of the population)
  - Members of the sample are chosen independently of each other
- If we take a simple random sample of a random value distribution, another way to say this is the elements of the sample are independent and identically distributed (i.i.d.)
  - Identically distributed means they're from the same distribution
  - Independence means that knowing one sample is included does not affect the probability of another sample being included

Law of Large Numbers

- Let X_1, X_2, …, X_n be independent and identically distributed random variables, each with mean μ. As the sample size n increases to infinity

$$\overline{X_n} = \frac{X_1 + X_2 + \ \ldots + X_n}{n} = \sum_{i=1}^{n} \frac{X_i}{n} \rightarrow \mu$$

- The sample mean $(\overline{X_n})$ converges to the random variable mean (μ)
  - In other words, for a large enough sample, the sample mean gets close to the random variable mean

LLN: Bernoulli example

- Suppose X_1, …, X_n are independent Bernoulli random variables
  - X_i = 1 with probability p (success, heads, etc.)
  - X_i = 0 with probability 1-p (failture, tails, etc.)
- Recall that the random variable mean = EX_i = 1*p + 0*(1-p) = p
- LLN says as n → infinity, the sample mean converges to the random variable mean:

$$\frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} \rightarrow p$$

LLN: 6-sided die example

- Suppose X_1, …, X_n are the numbers when we independently roll a fair die 6-sided die
- Random variable mean = EX_i = 1/6+2/6+3/6+4/6+5/6+6/6 = 3.5
- LLN says as n → infinity, the sample mean converges to the random variable mean:

$$\frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} \rightarrow 3.5$$

Central Limit Theorem

- X_1, …, X_n are independently and identically distributed random variables, each with mean μ and standard deviation σ. As sample size n increases to infinity:

$$\sum_{i=1}^{n} \frac{(X_i - \mu)}{\sqrt{n}\sigma} \Rightarrow Normal(0,1)$$

where Normal(0,1) is a normal random variable with mean 0 and standard deviation 1 (shortened N(0,1)).

Normal distribution

- For X = $\mathrm{N}(\mu, \sigma)$ and constants a and b:
  - aX+b is also a normal random variable with mean aμ+b and standard deviation aσ: $\mathrm{N}(\mathrm{a}\mu + b, a\sigma)$

Normal distribution

- Ubiquitous
- Sum of many independent variables results in a normal random variable
- E.g. sum of cells for height, measure of blood pressure, measure of cholesterol, batting averages

For large sample sizes n

$$\sum_{i=1}^{n} \frac{(X_i - \mu)}{\sqrt{n}\sigma} \approx N(0,1)$$

$$\sum_{i=1}^{n} (X_i - \mu) \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \mu \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i - n\mu \approx \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} X_i \approx N(n\mu, \sqrt{n}\sigma)$$

The **sum** of independent random variables is approximately **normally distributed** with mean $n\mu$ and standard deviation $\sqrt{n}\sigma$

For large sample sizes *n*,

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}}\, N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

The sample mean is also **normally distributed**, now the mean is μ and the standard deviation is $\sigma/\sqrt{n}$

For large sample sizes n,

$$\sum_{i=1}^{n} X_i \approx n\mu + \sqrt{n}\sigma N(0,1)$$

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}} N(0,1)$$

The Central Limit Theorem tells us **both** about the sum of independent random variables and their mean

The only difference is whether we divide by the sample size n

For large sample sizes *n*,

$$\sum_{i=1}^{n} \frac{X_i}{n} \approx \mu + \frac{\sigma}{\sqrt{n}} N(0,1)$$

The Law of Large Numbers tells us the sample mean is close to μ

The Central Limit Theorem tells us **how close**: $\frac{\sigma}{\sqrt{n}} N(0,1)$

Possible problems when applying the Central Limit Theorem

- Is an additive model appropriate?
- Are the random variables identically distributed? (baseball example)
- Are the random variables independent? (baseball example)
- How big does the sample size n have to be for the approximation to be valid?

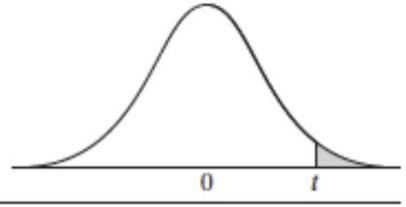
  n ~ > 20 or 30

Confidence intervals

- From Central Limit Theorem:
  - $\frac{\bar{Y}-\mu}{\sigma/\sqrt{n}}$ has a N(0,1) distribution
- Using our tables:
  - $P\left(-1.96 < \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} < 1.96\right) \approx .95$
- Isolating μ:
  - $P(\bar{Y} - (1.96\times \sigma/\sqrt{n}) < \mu < \bar{Y} + (1.96\times \sigma/\sqrt{n})) \approx .95$
- Notes:
  - The population mean, μ, is ***not*** random
  - The sample and confidence interval **are** random
  - We don't know σ

t-distribution

- If data is normally distributed, we can use it
- As df → ∞, t-distribution goes to normal distribution
- Each t value depends on df and confidence level
- Critical value – odds of getting a value as extreme **or** more extreme
- Two-sided:
  - 90% confidence: $t_{.05}$
  - 99% confidence: $t_{.005}$

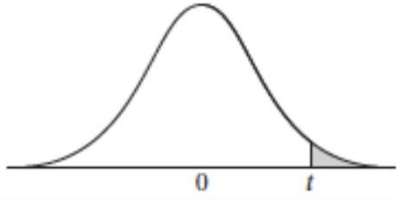


Critical Values of Student's *t* Distribution

	UPPER TAIL PROBABILITY									
df	0.20	0.10	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.888	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850
21	0.859	1.323	1.721	1.840	1.988	2.080	2.189	2.518	2.831	3.819
22	0.858	1.321	1.717	1.835	1.983	2.074	2.183	2.508	2.819	3.792
23	0.858	1.319	1.714	1.832	1.978	2.069	2.177	2.500	2.807	3.768
24	0.857	1.318	1.711	1.828	1.974	2.064	2.172	2.492	2.797	3.745
25	0.856	1.316	1.708	1.825	1.970	2.060	2.167	2.485	2.787	3.725
26	0.856	1.315	1.706	1.822	1.967	2.056	2.162	2.479	2.779	3.707
27	0.855	1.314	1.703	1.819	1.963	2.052	2.158	2.473	2.771	3.690
28	0.855	1.313	1.701	1.817	1.960	2.048	2.154	2.467	2.763	3.674
29	0.854	1.311	1.699	1.814	1.957	2.045	2.150	2.462	2.756	3.659
30	0.854	1.310	1.697	1.812	1.955	2.042	2.147	2.457	2.750	3.646
40	0.851	1.303	1.684	1.796	1.936	2.021	2.123	2.423	2.704	3.551
50	0.849	1.299	1.676	1.787	1.924	2.009	2.109	2.403	2.678	3.496
60	0.848	1.296	1.671	1.781	1.917	2.000	2.099	2.390	2.660	3.460
70	0.847	1.294	1.667	1.776	1.912	1.994	2.093	2.381	2.648	3.435
80	0.846	1.292	1.664	1.773	1.908	1.990	2.088	2.374	2.639	3.416
100	0.845	1.290	1.660	1.769	1.902	1.984	2.081	2.364	2.626	3.390
140	0.844	1.288	1.656	1.763	1.896	1.977	2.073	2.353	2.611	3.361
1000	0.842	1.282	1.646	1.752	1.883	1.962	2.056	2.330	2.581	3.300
∞	0.842	1.282	1.645	1.751	1.881	1.960	2.054	2.326	2.576	3.291
	60%	80%	90%	92%	94%	95%	96%	98%	99%	99.9%
	CRITICAL VALUE FOR CONFIDENCE LEVEL									

t-distribution

$\bar{Y} \pm (t \times s/\sqrt{n})$ where df = n-1



Critical Values of Student's *t* Distribution

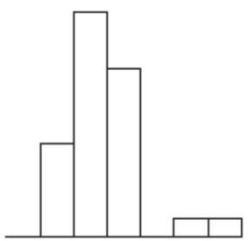
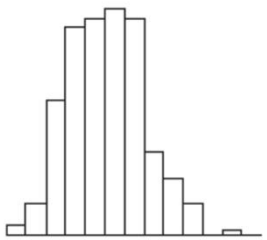
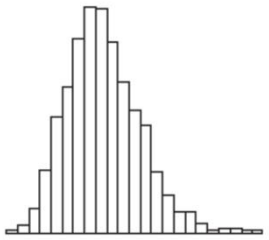
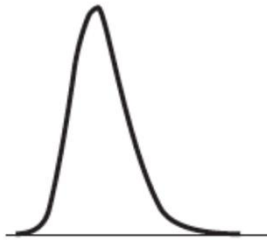
df	UPPER TAIL PROBABILITY									
	0.20	0.10	0.05	0.04	0.03	0.025	0.02	0.01	0.005	0.0005
1	1.376	3.078	6.314	7.916	10.579	12.706	15.895	31.821	63.657	636.619
2	1.061	1.886	2.920	3.320	3.896	4.303	4.849	6.965	9.925	31.599
3	0.978	1.638	2.353	2.605	2.951	3.182	3.482	4.541	5.841	12.924
4	0.941	1.533	2.132	2.333	2.601	2.776	2.999	3.747	4.604	8.610
5	0.920	1.476	2.015	2.191	2.422	2.571	2.757	3.365	4.032	6.869
6	0.906	1.440	1.943	2.104	2.313	2.447	2.612	3.143	3.707	5.959
7	0.896	1.415	1.895	2.046	2.241	2.365	2.517	2.998	3.499	5.408
8	0.889	1.397	1.860	2.004	2.189	2.306	2.449	2.896	3.355	5.041
9	0.883	1.383	1.833	1.973	2.150	2.262	2.398	2.821	3.250	4.781
10	0.879	1.372	1.812	1.948	2.120	2.228	2.359	2.764	3.169	4.587
11	0.876	1.363	1.796	1.928	2.096	2.201	2.328	2.718	3.106	4.437
12	0.873	1.356	1.782	1.912	2.076	2.179	2.303	2.681	3.055	4.318
13	0.870	1.350	1.771	1.899	2.060	2.160	2.282	2.650	3.012	4.221
14	0.868	1.345	1.761	1.888	2.046	2.145	2.264	2.624	2.977	4.140
15	0.866	1.341	1.753	1.878	2.034	2.131	2.249	2.602	2.947	4.073
16	0.865	1.337	1.746	1.869	2.024	2.120	2.235	2.583	2.921	4.015
17	0.863	1.333	1.740	1.862	2.015	2.110	2.224	2.567	2.898	3.965
18	0.862	1.330	1.734	1.855	2.007	2.101	2.214	2.552	2.878	3.922
19	0.861	1.328	1.729	1.850	2.000	2.093	2.205	2.539	2.861	3.883
20	0.860	1.325	1.725	1.844	1.994	2.086	2.197	2.528	2.845	3.850
21	0.859	1.323	1.721	1.840	1.988	2.080	2.189	2.518	2.831	3.819
22	0.858	1.321	1.717	1.835	1.983	2.074	2.183	2.508	2.819	3.792
23	0.858	1.319	1.714	1.832	1.978	2.069	2.177	2.500	2.807	3.768
24	0.857	1.318	1.711	1.828	1.974	2.064	2.172	2.492	2.797	3.745
25	0.856	1.316	1.708	1.825	1.970	2.060	2.167	2.485	2.787	3.725
26	0.856	1.315	1.706	1.822	1.967	2.056	2.162	2.479	2.779	3.707
27	0.855	1.314	1.703	1.819	1.963	2.052	2.158	2.473	2.771	3.690
28	0.855	1.313	1.701	1.817	1.960	2.048	2.154	2.467	2.763	3.674
29	0.854	1.311	1.699	1.814	1.957	2.045	2.150	2.462	2.756	3.659
30	0.854	1.310	1.697	1.812	1.955	2.042	2.147	2.457	2.750	3.646
40	0.851	1.303	1.684	1.796	1.936	2.021	2.123	2.423	2.704	3.551
50	0.849	1.299	1.676	1.787	1.924	2.009	2.109	2.403	2.678	3.496
60	0.848	1.296	1.671	1.781	1.917	2.000	2.099	2.390	2.660	3.460
70	0.847	1.294	1.667	1.776	1.912	1.994	2.093	2.381	2.648	3.435
80	0.846	1.292	1.664	1.773	1.908	1.990	2.088	2.374	2.639	3.416
100	0.845	1.290	1.660	1.769	1.902	1.984	2.081	2.364	2.626	3.390
140	0.844	1.288	1.656	1.763	1.896	1.977	2.073	2.353	2.611	3.361
1000	0.842	1.282	1.646	1.752	1.883	1.962	2.056	2.330	2.581	3.300
∞	0.842	1.282	1.645	1.751	1.881	1.960	2.054	2.326	2.576	3.291
	60%	80%	90%	92%	94%	95%	96%	98%	99%	99.9%
	CRITICAL VALUE FOR CONFIDENCE LEVEL									

Confidence interval

- A higher confidence interval is wider
  - 95% confidence interval is wider than 90%
  - 100% confidence interval would include everything
- Confidence interval is based on sample
- Sample is random so the confidence interval is random
  - Confidence interval only works if sample is from a **simple random sample**
  - If the sample size is small, the distribution **must** be approximately normal
  - If sample size is large (**n ~ 30**), distribution does not need to be normal (CLT)
    - But look for outliers regardless
- 95% confidence interval means if we repeat this many times, the **population mean** would be included in 95% of them
  - A particular calculated interval either has the true population mean or not
- As sample size increases, the width of the interval decreases

Standard error

- $SE_{\bar{Y}} = \frac{s}{\sqrt{n}}$
  - E.g. to lower SE by a factor of 2, we need 4 times the number of samples
- Measures **uncertainty** in our **estimate** of the population mean (μ)
- Different than standard deviation (SD) which measures the dispersion of the data

	$n = 28$	$n = 280$	$n = 2{,}800$	$n \to \infty$
$\bar{y}$	5.17	5.19	5.14	$\bar{y} \to \mu$
s	0.65	0.67	0.65	$s \to \sigma$
SE	0.12	0.040	0.012	SE $\to 0$
Sample distribution				

One-sided confidence intervals

- t
  - We don’t need the factor of 2
  - E.g. for a 90% confidence interval we use $t_{0.1}$
- Lower bound:
  - $\bar{Y} - (t \times s/\sqrt{n})$
  - Confidence interval: $(\bar{Y} - (t \times s/\sqrt{n}), \infty)$
- Upper bound:
  - $\bar{Y} + (t \times s/\sqrt{n})$
  - Confidence interval:$(-\infty, \bar{Y} + (t \times s/\sqrt{n}))$

Comparing two means

- Requires 2 **independent** samples
- We can compare μ_1 and μ_2:
  - $\mu_1 - \mu_2$, is it different than 0?
  - This is the same as $\mu_1 = \mu_2$
- Standard error of each distribution:
  - $SE_{\bar{Y}_1} = \frac{s_1}{\sqrt{n_1}}$
  - $SE_{\bar{Y}_2} = \frac{s_2}{\sqrt{n_2}}$
- Standard error of the difference:
  - $SE_{\bar{Y}_1 - \bar{Y}_2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Confidence interval for $\mu_1 - \mu_2$

- Requires 2 samples to be independent and normal
  - $(\bar{y}_1 - \bar{y}_2) \pm (t \times SE_{(\bar{Y}_1 - \bar{Y}_2)})$
- Degrees of freedom
  - $df = \frac{(SE_1^2 + SE_2^2)^2}{SE_1^4/(n_1-1) + SE_2^4/(n_2-1)}$
- For textbook problems, it will be computed for you
  - Round down to nearest calculated df
- R can compute this automatically
- Otherwise, we can use 2 approximations:
  - **Conservative approximation: min((n_1-1), (n_2-1))**
    - **Use this one**
  - Liberal approximation: n_1+n_2-2